

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Asbestos: fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma; disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) • Cyanide: very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested; disposal: incineration so carbon and nitrogen are separated and oxidised to CO₂ and NO_x (much smaller problem) • High-level radioactive waste: intense ionising radiation, heat generation, long half-lives; disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores	6
01.2	• The fibres are bound / cannot become airborne so cannot be inhaled	1
01.3	• Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material • The solid cement immobilises the waste so it cannot leak / leach out	2
01.4	• Even if the glass shatters the waste stays encapsulated in the fragments; glass is more stable / resistant to heat and leaching over very long timescales	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
02.1	• Terracing: sloping land divided into narrow level fields held by retaining walls along contours; water slows as it crosses each terrace and infiltrates instead of running off • Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff • Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles; protect bare fields after harvest • Mulching: organic matter on the surface protects soil from raindrop impact and wind; decomposes to humus which binds particles; retains moisture • Strip cropping: strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field; slows runoff	5
02.2	• No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles • Permanent root systems and canopy cover protect the soil all year	2
02.3	<i>Any 3 from:</i> • Control stocking density so vegetation regrows faster than it is eaten • Restrict access to vulnerable areas such as river banks (fencing) • Move / remove livestock at high-risk times, eg after heavy rain when soil is wet • Rotational grazing to rest pastures	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	
03.1	• Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron • A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) • The heat boils water to steam which drives turbines and generators	3
03.2	• Output changes slowly (days) so it cannot follow rapid fluctuations in demand • Very high construction cost but low fuel cost means it is most economic running continuously at full output	2
03.3	• Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal • Still need condenser cooling water (coast / large lake) and a stable site	2
03.4	• Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy; reactor construction and decommissioning have embodied energy	1
03.5	<i>Any 2 from:</i> • No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates • Large projects often overrun original cost estimates • Long-term waste disposal costs uncertain • State subsidies may hide costs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
04.1	The time taken for half of the atoms of a radioactive isotope to decay (activity halves)	1
04.2	$32 / 8 = 4$ half-lives $6400 / 2^4 = 400$ Bq	2
04.3	- Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient - Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low - Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations)	3
04.4	- Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure - Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) - Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure)	3
04.5	A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
05.1	- The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter - Current recharge is little or none so any abstraction reduces the stored volume - The large volume means shortages take a long time to appear, but eventually the supply runs out; ancient water is often saline, causing soil salinisation	3
05.2	- Two satellites whose orbits are affected by the force of gravity - The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass - Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas	3
05.3	<i>Any 3 from:</i> - Low-pressure irrigation pipes to reduce leaks - Growing crops that need less water than wheat / maize - Charging consumers by volume used (metering) - Inter-basin transfer of water 450 km from the Yangtze	3
05.4	Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
06.1	- Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) - Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect)	2

Question	Marking guidance	Marks
06.2	• $(930 - 250) / 250 \times 100$ • = 272%	2
06.3	• Anticyclonic conditions: clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled • Low wind velocity so air layers did not mix; London lies in a valley / basin where cold dense air collected: a temperature inversion formed • Smoke from coal fires was trapped below the inversion; moist air cooled to its dew point forming fog; smoke + fog = smog; the smog's high albedo prolonged the inversion	3
06.4	• Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections • These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal	2
06.5	• Clean Air Act (1956)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
07.1	• Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed; hydrodynamics for ships • Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption • Conventional friction brakes convert kinetic energy to waste heat; regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again	6
07.2	• Pneumatic tyres deform as they rotate; the continual change in shape and air movement inside generates frictional heat (more if under-inflated) • Steel wheels do not deform so rolling resistance is very low	2
07.3	<i>Any 2 from:</i> • More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel • Better temperature control (thermostatic fan) • Electronic ignition control / regular servicing to keep spark timing correct • Turbochargers supplying more oxygen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
08.1	• $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ • $N(N - 1) = 100 \times 99 = 9900$ • $D = 9900 / 2900 = 3.41$ (accept 3.4)	3
08.2	• Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
09.1	• Temperature inversion	1
09.2	• On night A, temperature falls with altitude; warm effluent gases are less dense / more buoyant than the surrounding air so they rise • As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting • On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) • Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated	4
09.3	<i>Any 3 from:</i> • Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly • The cold ground cools the air in contact with it, producing a cold dense layer near the ground • Low wind velocity means the air layers of different temperatures do not mix • Valley allows cold dense air to collect / drain into the low ground	3
09.4	<i>Any 2 from:</i> • Smoke smogs (eg London 1952) • Photochemical smogs (eg Los Angeles, Mexico City) • Build-up of NO_x / carbon monoxide / PM₁₀ in cities	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
10.2	• All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25

Question	Marking guidance	Marks
11.2	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25